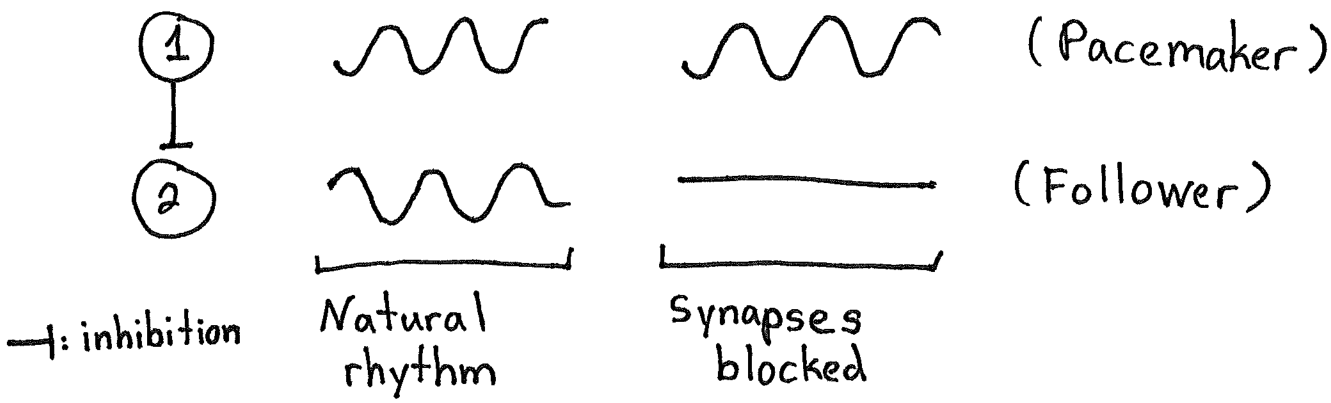
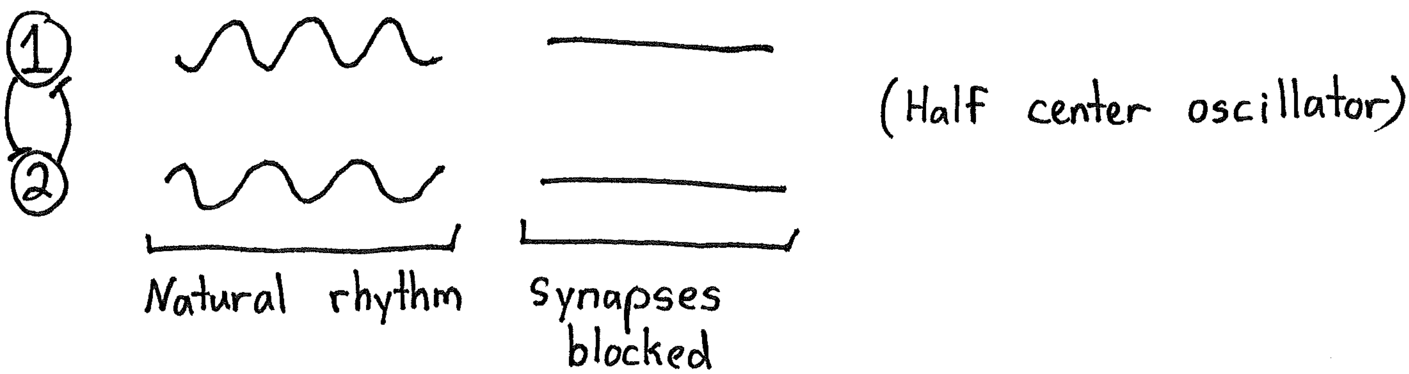
Central Pattern Generators:

As mentioned in motor systems introduction, rhythmic motions of the body often follow rhythmic oscillations of neurons. Some rhythms are generated cell intrinsically.



A well-known example is respiration.

Other rhythms are generated by the network itself. The basic circuit motif is mutual inhibition

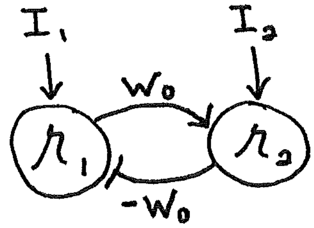


A well-known example is muscle alternation during walking.

We want to understand the neural mechanisms behind both rhythm types. We'll see that nonlinearity is essential to get realistic patterns.

Oscillations: Central Pattern Generators (Linear)

Let's consider a different two-neuron example.



$\longrightarrow$: excitation

$\dashv$: inhibition ; $W = \begin{bmatrix} 0 & -W_0 \\ W_0 & 0 \end{bmatrix}$

The first step of our program is to find eigenvalues

$$0 = \det \begin{bmatrix} \lambda & W_0 \\ -W_0 & \lambda \end{bmatrix} = \lambda^2 + W_0^2 \Rightarrow \lambda_\pm = \pm i \sqrt{W_0^2} = \pm i |W_0|$$

The eigenvalues are imaginary! Suspending disbelief,

$$\tau \frac{d\alpha_\pm}{dt} = -(1 \mp i|W_0|)\alpha_\pm + \beta_\pm$$

$$\tau_{EFF,\pm} = \frac{\tau}{(1 \mp i|W_0|)} \Rightarrow e^{-t/\tau_{EFF,\pm}} = e^{-t(1 \mp i|W_0|)/\tau}$$

$$= \underbrace{e^{-t/\tau}}_{\substack{\text{exponential} \\ \text{decay} \\ \text{(real part)}}} \underbrace{e^{\pm i \frac{|W_0|}{\tau} t}}_{\substack{\text{What's} \\ \text{this?} \\ \text{(imaginary part)}}}$$

Fact about complex numbers:

$$e^{i\theta} = \cos(\theta) + i \sin(\theta)$$

$$\Rightarrow e^{\pm i \frac{|W_0|}{\tau} t} = \underbrace{\cos\left(\frac{|W_0|}{\tau} t\right) \pm i \sin\left(\frac{|W_0|}{\tau} t\right)}_{\text{oscillations!}}$$

When the weight matrix has complex eigenvalues, this means that there are oscillatory components to the dynamics, but the math becomes a bit cumbersome, and we won't spend more time on analyzing oscillations analytically in linear systems.

Nonlinear oscillations:

Pacemaker neurons:

The Morris-Lecar model is a simple Hodgkin-Huxley-like model that is capable of spontaneous oscillations.

$$C\frac{dV}{dt} = -g_L(V-E_L) - \underbrace{\bar{g}_{Ca} M_\infty(V)(V-E_{Ca})}_{\text{Depolarizing calcium current}}$$

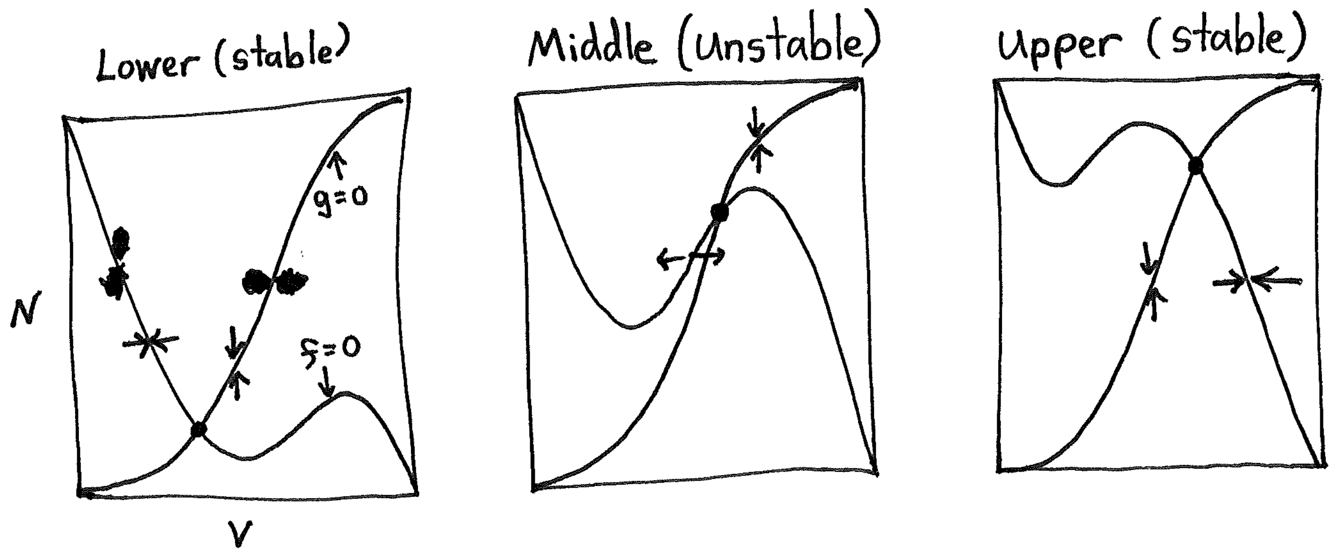
$$\underbrace{-\bar{g}_K N(V-E_K)}_{\text{Hyperpolarizing potassium current}} + \underbrace{I_e}_{\text{External input}}$$

$$\tau_N \frac{dN}{dt} = -N + N_\infty(V)$$

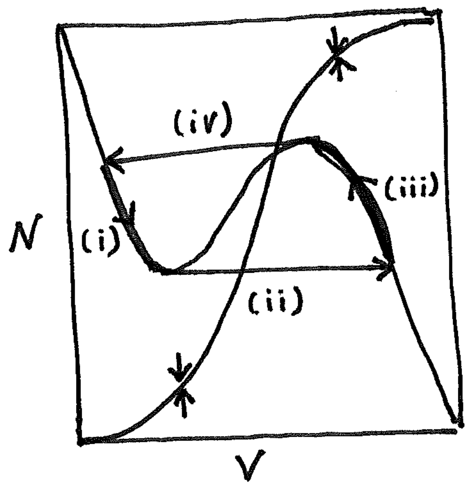
Both M_∞ and N_∞ are sigmoidal curves. We will analyze the dynamics geometrically. First, we have

$$\frac{dV}{dt} = f(V,N) \quad , \quad \frac{dN}{dt} = g(V,N)$$

in abstract form, with V-nullcline $f(V,N)=0$ and N-nullcline $g(V,N)=0$. We plot them and find three fixed point types are possible because of "cubic-like" V-nullcline



The dynamics are oscillatory when the fixed point is unstable. This is easiest seen by noting that the V-dynamics are faster than the N-dynamics. The system thus relaxes quickly to the V-nullcline and follows it in the direction of dN/dt.



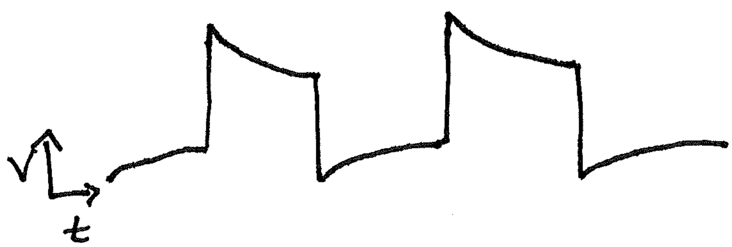
(i) V increases as N decreases

(ii) V jumps from end of lower branch to upper branch. (It can't go further as N decreases)

(iii) V decreases as N increases

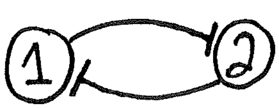
(iv) V jumps from end of upper branch to lower branch.

As this cycle repeats, V oscillates autonomously



Half-center oscillators:

Two inhibitorially coupled Morris-Lecar neurons can oscillate even if the single neurons do not.



$$C\frac{dV_1}{dt} = C f(V_1, N_1) - \bar{g}_s \underbrace{S_\infty(V_2)}_{\text{sigmoid}}(V_1 - \underbrace{E_s}_{\sim -80\text{mV}})$$

$$C\frac{dV_2}{dt} = C f(V_2, N_2) - \bar{g}_s S_\infty(V_1)(V_2 - E_s)$$

$$\frac{dN_i}{dt} = g(V_i, N_i)$$

The magnitude of the synaptic conductance is a sigmoidal function of the pre-synaptic voltage. For simplicity, we model it with a sharp threshold

$$S_\infty(V) = \begin{cases} 0, & V < V_T \\ 1, & V > V_T \end{cases}$$

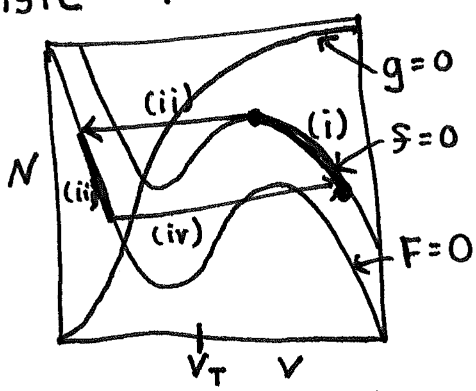
When the presynaptic neuron's voltage is below the synaptic threshold, V_T, the postsynaptic neuron evolves freely. When above, the postsynaptic neuron is inhibited

$$\underbrace{I_s}_{\text{synaptic current}} = -\bar{g}_s (V - E_s)$$

$$\frac{dV}{dt} = f(V, N) - \frac{\bar{g}_s}{C}(V - E_s) = F(V, N)$$

In a half-center oscillator, the two neurons alternate between being free and inhibited. The free neuron's nullclines are $f(V,N) = 0$, $g(V,N) = 0$. The inhibited neuron's nullclines are $F(V,N) = 0$, $g(V,N) = 0$.

Intrinsic release of inhibition mechanism



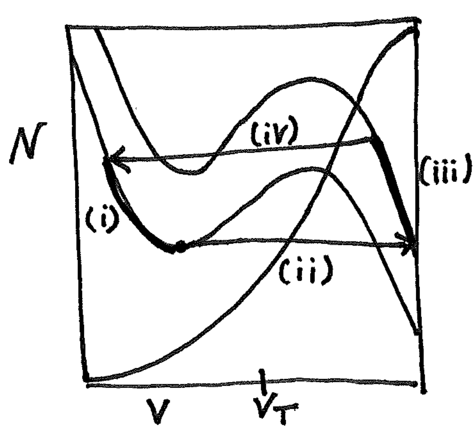
(i) Free neuron decreases V as N increases

(ii) Free neuron jumps to low voltage and releases inhibition at end of upper branch. Becomes inhibited.

(iii) Inhibited neuron increases V as N decreases

(iv) Inhibited neuron becomes free and jumps to f-nullcline.

Intrinsic escape from inhibition mechanism:



(i) Inbibited increases V as N decreases

(ii) Inhibited neuron jumps to high voltage and escapes inhibition at end of lower branch. Becomes free.

(iii) Free neuron decreases V as N increases.

(iv) Free neuron becomes inhibited and jumps to F-nullcline

Half-center rate networks:

The same geometry of nullclines can produce half-center oscillators in firing rate models. Replace V with r and N with an adaptation/recovery variable.